Resolution refinement and the rodeo algorithm beyond seniority: eigenstate preparation for the pairing Hamiltonian with a particle-hole interaction

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The constant-pairing Hamiltonian conserves seniority, so its ground state lives in the seniority-zero subspace, the one-particle–one-hole channel is closed, and a one-qubit-per-pair encoding is available — properties that make it a convenient but unusually well-behaved testbed for quantum eigenstate preparation. We remove this simplification by adding to the pairing force a seniority-breaking particle-hole interaction of strength f, which carries the one-particle–one-hole and pair-breaking channels of realistic two-body interactions, and we repeat on the resulting Hamiltonian the assessment of a companion paper: resolution refinement — the adiabatic transport of an eigenstate from a coarse to a fine single-particle basis along a shift-controlled, gapped path — followed by the rodeo algorithm as a projective energy filter, compared against the same filter applied to a full-space unitary coupled-cluster (UCCSD) state, with the ground-state infidelity as the common observable and a two-qubit gate budget per accepted preparation. For four particles on four levels (eight system qubits) at $f = 0.05$, 0.2 and 0.5 the structure, operating point and convergence of the pipeline are unchanged: the path gap is still set by the energy shift, the refinement time grows from 16 to 32 only because the prolonged starting state loses overlap, and is governed by the renormalized pairing strength $g + 2f$; the filter converges geometrically to an infidelity of 4.5×10^{-5} after ten cycles at the strongest pair breaking. The price of the more realistic interaction appears where it should, in the Pauli-string content of the Hamiltonian ($61 \to 325$ strings on eight qubits, maximal weight $4 \to 8$) and hence in the depth of a Trotter step ($296 \to 2728$ CNOTs), which raises the budget per accepted preparation from 7×10^4 to 6–8×10^5 CNOTs. The single-reference UCCSD ansatz, exact for the pairing force at the coarse resolution and accurate to 1.8×10^{-4} in infidelity at the fine one, degrades twentyfold across the scan as the ground state acquires seniority-two weight, yet filtering it still reaches a lower infidelity than the refined input at a factor two to three lower cost. The sufficient adiabatic-time and Trotter-step conditions available for such paths are evaluated and shown to be insensitive to the pair breaking in the time and sensitive to it in the step count.

I. INTRODUCTION

Schematic nuclear Hamiltonians — pairing, Lipkin and small shell-model interactions — are the standard proving ground for quantum algorithms in nuclear structure [2–11], and the constant-pairing model in particular, integrable by the Richardson ansatz [12–14], has become the model of choice: the programme of Lacroix and collaborators on symmetry breaking and restoration [15–20], on ADAPT-VQE with symmetry-conserving and symmetry-breaking pools [10, 21] and on Green's functions of small superfluid systems [22] is built largely on it, and so is the companion paper to the present work [1], in which we implemented and assessed, stage by stage against exact results, a pipeline for eigenstate preparation that combines *resolution refinement* [23] — the adiabatic transport of a coarse-resolution eigenstate, lifted by a prolongation operator into a finer single-particle basis, along a shift-controlled gapped path — with the *rodeo algorithm* [24, 25] as a projective energy filter. The finding of that assessment was, on the question that decides whether the refinement stage earns its gates, negative: compared on the same observable, the ground-state infidelity, and with the same Trotterized filter, refinement followed by the rodeo algorithm does not outperform the filter applied to a full-space unitary coupled-cluster (UCCSD) state at any size that can be validated exactly; what the refinement offers in exchange is freedom from classical optimization and training measurements.

The pairing Hamiltonian is, for the purpose of testing methods, almost too well behaved. The pairing force moves complete time-reversed pairs and does noth-

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ing else: it conserves the seniority (the number of unpaired particles), the $N = 4$ ground state on k levels lives in a $\binom{k}{2}$-dimensional seniority-zero space, and an entire class of excitations is absent — there is no matrix element between the reference determinant and any one-particle–one-hole state. The model is by no means outside the reach of mean-field-based methods — the Richardson model has long served as a testing ground for particle-particle RPA and its self-consistent and renormalized variants [26, 27] and for the quasiparticle RPA [28] — but the particle-hole channel on which Tamm–Dancoff and particle-hole RPA are built is simply closed in it. Realistic nuclear and electronic Hamiltonians do have that channel. Two consequences follow for quantum state preparation. A single-reference variational ansatz is unusually accurate for the pure pairing force, because the singles amplitudes vanish identically and the ansatz is effectively a doubles theory; and the compact one-qubit-per-pair encoding used in much of the pairing-model literature [15, 22] is available only because the seniority is conserved. Neither property survives a seniority-breaking interaction, and an assessment carried out on the pairing model alone is therefore silent on precisely the regime — a degrading variational input, a full fermionic register — in which an optimizer-free preparation such as refinement might be expected to pay off.

In this paper we open the particle-hole channel. Following Chapter 4 of Ref. [29] we add to the pairing force a single extra term, a particle-hole interaction $V_{\rm ph}$ of strength f that contains, besides a renormalization of the pairing strength, one-particle–one-hole excitations of seniority two and seniority-two two-particle–two-hole excitations, and we keep everything else — the level scheme, the particle number $N = 4$, the resolutions $k = 2, 3, 4$, the $2k$-qubit Jordan–Wigner register, and every stage and setting of the pipeline — exactly as in Ref. [1], so that each change in the results can be attributed to the interaction alone. The strengths $f = 0.05$, 0.2 and 0.5 at $g = 1$ are those of Ref. [29], whose tables every exact number below reproduces. We ask five questions: whether the pipeline runs unchanged on a Hamiltonian with a denser, seniority-mixed spectrum; how the refinement time and the path gap respond to the pair breaking; how the rodeo filter behaves when its energy scan sees a denser spectrum; what the extra channels cost at the circuit level, in Pauli strings, Trotter depth and two-qubit gates per accepted preparation; and whether the verdict on the refinement stage against the filtered UCCSD input changes once the variational input itself starts to degrade.

The paper is organized as follows. Section II defines the Hamiltonian and its qubit encoding. Section III recalls the stages of the pipeline, the cost accounting and the pure-pairing reference numbers of Ref. [1] that the results are measured against. Section IV presents the exact and variational references, the refinement, the sufficient adiabatic conditions and the rodeo filtering with the particle-hole term, and Sec. V the circuit-level budget. Section VI discusses the findings, the relation to the pairing-model literature, the hardware requirements and the extensions that follow.

II. THE PAIRING MODEL WITH A PARTICLE-HOLE INTERACTION

The constant-pairing Hamiltonian of Ref. [1] is

$$H_{\rm pair} = \delta \sum_{p=0}^{k-1} p \sum_{\sigma=\uparrow\downarrow} n_{p\sigma} - \frac{g}{2} \sum_{p,q=0}^{k-1} a^\dagger_{p\uparrow} a^\dagger_{p\downarrow} a_{q\downarrow} a_{q\uparrow}, \tag{1}$$

with k doubly degenerate single-particle levels of spacing δ (set to $\delta = 1$) and a pairing strength g; level p and spin σ are mapped to qubit $2p+\sigma$, so a k-level problem uses $2k$ qubits, and the fermionic operators are encoded through the Jordan–Wigner transformation. "Basis refinement" means increasing k at fixed N. To this we add, following Chapter 4 of Ref. [29], the particle-hole interaction

$$V_{\rm ph} = -\frac{f}{2} \sum_{p,q,r=0}^{k-1} \Big(a^\dagger_{p\uparrow} a^\dagger_{p\downarrow} a_{q\downarrow} a_{r\uparrow} + \text{h.c.} \Big), \tag{2}$$

and study $H = H_{\rm pair} + V_{\rm ph}$ at $g = 1$ and $f = 0.05$, 0.2, 0.5, with $f = 0$ the pure pairing model as the reference.

The triple sum contains three distinct processes. The $q = r$ pieces are the pairing term again and merely renormalize $g \to g + 2f$; the $q = p \neq r$ pieces carry a *single* particle from one level to another, a $1p$–$1h$ excitation of seniority two; and the pieces with p, q, r all different remove two unpaired particles and deposit a pair elsewhere, a seniority-two $2p$–$2h$ excitation. $V_{\rm ph}$ still conserves $N_\uparrow$ and $N_\downarrow$ separately, so S_z remains a good quantum number, but the seniority does not: at $k = 4$ the $S_z = 0$ sector has 36 determinants instead of the six pair configurations, and the compact one-qubit-per-pair encoding of the pairing model [15, 22] is no longer available. We keep the $2k$-qubit Jordan–Wigner register; it is the convenient general choice, not the only one — the conserved particle number and S_z still permit compressed encodings of the 36-dimensional block — but it is the register on which the gate counts below are meaningful, and it is the register on which the stages of the pipeline were written. Acting once on the reference determinant at $k = 4$, the Hamilto-

TABLE I. Pauli-string content of the Hamiltonian on $2k$ qubits: number of strings and largest Pauli weight, for the pairing model and for the pairing model with the particle-hole term ($f \neq 0$; the counts do not depend on the value of f). The $k = 2, 3$ rows agree with Ref. [29].

k	qubits	pairing	pairing + $V_{\rm ph}$
2	4	15 (weight 4)	27 (weight 4)
3	6	34 (weight 4)	118 (weight 6)
4	8	61 (weight 4)	325 (weight 8)

nian now reaches eight $1p$–$1h$ and eight seniority-two $2p$–$2h$ determinants with matrix elements $\pm f/2$, in addition to the four seniority-zero $2p$–$2h$ states whose coupling grows from $g/2$ to $(g+2f)/2$. Two consequences matter for the pipeline. First, the filled determinant is no longer a stationary Hartree–Fock solution: its one-particle–one-hole matrix elements $\langle\Phi_i^a|H|\Phi_0\rangle = \pm f/2$ do not vanish (Brillouin's theorem holds for the self-consistent determinant, which we have not re-optimized), so the UCCSD *singles*, identically zero for the pairing force, switch on and absorb part of what an optimized reference would describe as mean-field relaxation rather than correlation. Second, the reference energy becomes $E_{\rm ref} = 2 - g - 2f$, and the quantity we call correlation energy below is measured from this unrelaxed reference.

Table I lists the Pauli-string content of H on $2k$ qubits. The particle-hole term multiplies the number of strings by a factor that grows with the resolution (1.8, 3.5 and 5.3 at $k = 2, 3, 4$; $61 \to 325$ at $k = 4$) and raises the largest Pauli weight from four to $2k$ (8 at $k = 4$), because a $1p$–$1h$ string drags a Jordan–Wigner parity chain across the register. This table is the origin of every increase in circuit depth reported in Sec. V.

III. THE PIPELINE, ITS COST ACCOUNTING, AND THE PURE-PAIRING REFERENCE

We recall the stages of the pipeline and the conventions of Ref. [1] to the extent needed to read the results; the derivations, the circuit diagrams and the validation of every stage against exact results are given there.

A. Stages

a. Coarse state. The coarse ($k_{\rm low}$) eigenstate is prepared with a gate-level unitary coupled-cluster ansatz restricted to singles and doubles (UCCSD) [30, 31], Trotterized into a product of excitation circuits acting on the Hartree–Fock (HF) reference and optimized with its exact analytic gradient. Each excitation compiles, through the Jordan–Wigner map, into Pauli rotations $e^{-i\frac{\phi}{2}P}$, a string of weight w costing $2(w-1)$ CNOTs [32]; a pair-hopping double costs 48 CNOTs independent of the level separation, and the full $k = 4$, $N = 4$ preparation 26 operators and 1 312 CNOTs. At $N = 4$ the $k = 2$ sector is a single determinant, so the $2 \to 4$ chain used throughout has a gate-free coarse stage; the $3 \to 4$ chain starts from the 272-CNOT gate-level UCCSD state at $k = 3$. The same ansatz applied at the fine resolution is the *comparator* of the assessment.

b. Prolongation and refinement. The prolongation P maps every basis state of the $k_{\rm low}$ space to the same occupation pattern in the $k_{\rm high}$ space, leaving the new levels empty; in canonical Jordan–Wigner ordering it is gate-free. The prolonged state $|\Phi_0\rangle = P\,|\psi_{\rm low}\rangle$ is refined by adiabatically following the interpolated, energy-shifted Hamiltonian

$$H(s) = \cos^2\!\Big(\tfrac{\pi s}{2}\Big)\,P(H_{\rm low}-\mu)P^\dagger + \sin^2\!\Big(\tfrac{\pi s}{2}\Big)\,(H_{\rm high}-\mu), \qquad s = \frac{t}{T}, \tag{3}$$

evolved by a Suzuki–Trotter product over total time T. The shift $\mu = E_{\rm low} + 0.6$ places the tracked low-energy states below the large null space of $P^\dagger$; since $PP^\dagger$ projects on the states in which the appended levels are empty, the first term is a coarse-space operator conditioned on the $2(k_{\rm high} - k_{\rm low})$ new qubits being in $|0\rangle$,

$$A \equiv P(H_{\rm low}-\mu)P^\dagger = (H_{\rm low}-\mu)\otimes|0\cdots0\rangle\langle0\cdots0|_{\rm app}\,, \tag{4}$$

whose circuit is not free (Sec. III B). At $s = 0$ the gap above the tracked state is $\mu - E_{\rm low}$ for a one-dimensional coarse space, and along the path the instantaneous spectrum was found to stay smooth and gapped in every case of Ref. [1]. The figure of merit of the stage is the overlap $p = |\langle\Psi_{\rm high}|\Phi(T)\rangle|^2$, and T^* denotes the time at which it reaches 0.99 on the discrete T grid of Ref. [1] (from $T = 0.5$ to 80). We call "exact-slice" the refinement in which every slice of the path is applied as the exact exponential within the N-particle sector, and "Trotterized" the product-formula version used at the circuit level (Sec. V). The $\sin^2(\pi s/2)$ schedule is the one for which the adiabatic bound of Jansen, Ruskai and Seiler [33] loses its boundary terms, and Ref. [34] turns that bound into explicit sufficient conditions for a Trotterized adiabatic

stage: a total time

$$T \;\geq\; \frac{1}{\sqrt{\epsilon}} \int_0^1 \left[\frac{\left\| \ddot{H} \right\|}{\Delta^2} + 7\, \frac{\left\| \dot{H} \right\|^2}{\Delta^3} \right] ds \tag{5}$$

guarantees a final infidelity at most ϵ (derivatives with respect to s, $\Delta(s)$ the instantaneous gap), and $n_s \geq \eta^{-1}(T\Lambda_1/4 + T^2\Lambda_2/2)$ first-order Trotter steps keep the operator error of the digitized evolution below η, with $\Lambda_1 = \max_s \left\| \dot{H} \right\|$ and Λ_2 the largest commutator norm between the groups into which the Hamiltonian is split. Both conditions are sufficient, not necessary [34, 35]; the refinement times T^* reported below are empirical, and Sec. IV C evaluates the conditions for comparison.

c. Rodeo filter. One cycle of the rodeo algorithm [24, 25] acts on the system register plus one ancilla: a Hadamard, the controlled evolution e^{-iHt_k}, the ancilla phase $P(Et_k)$, a second Hadamard and an ancilla measurement; post-selecting $|0\rangle$ applies the filter $\frac{1}{2}(\mathbb{1} + e^{iEt_k} e^{-iHt_k})$, so that an eigenstate $|E_j\rangle$ acquires the factor $\cos[(E_j - E)t_k/2]$ in modulus. With the t_k drawn from a Gaussian of RMS width σ the Gaussian-averaged single-cycle factor is $q_k = \frac{1}{2}\big(1 + e^{-(E_k - E)^2\sigma^2/2}\big)$, so for an input $|\psi\rangle = \sum_k c_k |E_k\rangle$ and $E = E_0$ the acceptance probability after M cycles tends to $p = |c_0|^2$ and the acceptance-weighted fidelity of the ensemble of accepted runs obeys exactly

$$1 - \mathcal{F}_M = \frac{\sum_{k\neq 0} |c_k|^2 q_k^M}{p + \sum_{k\neq 0} |c_k|^2 q_k^M} \leq \frac{(1-p)\, q_{\max}^M}{p + (1-p)\, q_{\max}^M}, \tag{6}$$

which for $\Delta_{\min}\sigma \gg 1$ reduces to the familiar $\frac{1-p}{p} 2^{-M}$. Scanning E locates eigenvalues as peaks of the all-zero probability above the 2^{-M} background of far-off-resonant components; fixing E at the located ground-state energy turns the algorithm into a state-preparation filter. In Sec. IV the controlled evolution is applied exactly within the N-particle sector; in Sec. V it is a product formula with a fixed maximum step size δt, a cycle of Gaussian time t_k using $n_k = \lceil |t_k|/\delta t \rceil$ steps.

B. Circuit compilation and cost accounting

All operations are compiled to single-qubit rotations and CNOTs and simulated block by block: each term of the Hamiltonian splitting is a set of mutually commuting Pauli strings whose exponential is realized exactly by its rotation–CNOT block, applied within the conserved N-particle sector, so the product-formula error comes only from the non-commutativity between blocks, exactly as on a device, and the CNOT counts are those of the compiled blocks. For the pairing force the splitting has one diagonal block D (kinetic energy plus the $p = q$ pairing term) and one pair-hopping block h_{pq} per level pair — seven blocks at $k = 4$ — and a first-order Trotter step contains the $k(k-1)/2$ pair-hopping terms (8 weight-4 rotations, 48 CNOTs each), k diagonal ZZ rotations (2 CNOTs each) and $2k$ single-qubit phases, i.e.

$$N_{\mathrm{CNOT}}^{\mathrm{step}}(k) = 24\, k(k-1) + 2k \tag{7}$$

CNOTs uncontrolled — 296 at $k = 4$ — and 416 once the central R_z of every rotation is controlled for the rodeo cycle (2 extra CNOTs per rotation). The conditioned coarse term of Eq. (4) is, for the $2 \to 4$ chain at $N = 4$, a phase conditioned on the four appended qubits being empty, compiled along a Gray code into 14 CNOTs and 15 rotations per application; it is unchanged by V_{ph}, since the coarse space is still one determinant. The two-qubit gates per accepted preparation are

$$C_M = \frac{C_{\mathrm{prep}} + \sum_{m=1}^{M} \langle P_{m-1} \rangle\, c_m}{P_M}, \tag{8}$$

with C_{prep} the cost of the input (for the refinement, $2n_s$ second-order half-steps of $N_{\mathrm{CNOT}}^{\mathrm{step}} + 14$ CNOTs), c_m the cost of the m-th controlled cycle, $\langle P_{m-1} \rangle$ the probability of reaching it, and P_M the acceptance; one-time costs — the UCCSD training and the energy scan — are reported separately. The *operating point* at which all inputs are compared is $n_s = 80$ second-order refinement steps along the $T = 30$ path, $M = 2$ rodeo cycles with $\sigma = 4$, $\delta t = 0.25$, and 40 common Gaussian time sets.

C. The pure-pairing reference

The numbers of Ref. [1] at $k = 4$, $N = 4$, $g = 1$ against which everything below is measured are the following. The physical gap of H_{high} is 1.823 and the minimum gap along the $2 \to 4$ path is the shift-controlled 0.600. The prolonged determinant has overlap $\mathrm{ov}_0 = 0.868$ with the exact ground state; exact-slice refinement reaches 0.99 at $T^* = 16$ and 0.9988 at $T = 30$, which $n_s = 80$ second-order steps reproduce as 0.9977. The rodeo filter with exact controlled evolution ($\sigma = 3$) brings the refined $T = 25$ input ($p = 0.9976$) to an infidelity of 6.5×10^{-4} after two cycles and 2×10^{-6} after ten. The $k = 4$ UCCSD state has an energy error of 1.4×10^{-3} but an actual infidelity of only 1.78×10^{-4} — an energy error is not an infidelity, since $\langle H \rangle - E_0 \geq \Delta\,(1 - \mathcal{F})$ — and at the operating point the Trotterized filter brings it to 1.3×10^{-4} for 2.3×10^4 CNOTs per accepted preparation, whereas the refined $T = 30$ input reaches 5.3×10^{-4} for 7.2×10^4; all inputs converge after six cycles to the floor 1.1–1.5×10^{-4} set by the product formula at $\delta t = 0.25$. A too-coarse step is rejected by the filter itself: at $\delta t = 1$ the acceptance collapses to 0.10, the diagnostic we shall meet again at $f = 0.5$. Finally, the sufficient time condition of Eq. (5) demands $T \geq 5.1 \times 10^3$ for an infidelity of 10^{-2} against the empirical $T^* = 16$, and the step condition 3.4×10^4 first-order steps at T^* against 160 in practice.

TABLE II. The pairing model with the particle-hole term of Eq. (2), $N = 4$, $g = 1$. $E_{\mathrm{ref}} = 1 - 2f$ is the energy of the reference determinant, FCI the exact N-sector ground state, UCCSD the statevector unitary coupled-cluster energy (identical to the gate-level circuit at $k = 3$), $\Delta E_{\mathrm{UCCSD}} = E_{\mathrm{UCCSD}} - E_{\mathrm{FCI}}$, $1 - \mathcal{F}_{\mathrm{UCCSD}}$ the infidelity of the optimized UCCSD state with respect to the exact ground state, and w_0 the weight of the exact ground state in the seniority-zero subspace. The $k = 4$ FCI column agrees with the corresponding table of Ref. [29].

k	f	FCI	UCCSD	$\Delta E_{\mathrm{UCCSD}}$	$1 - \mathcal{F}_{\mathrm{UCCSD}}$	w_0
3	0	0.794697	0.794697	$< 10^{-14}$	$< 10^{-14}$	1
3	0.05	0.650378	0.650378	$< 10^{-14}$	$< 10^{-14}$	0.9994
3	0.2	0.185803	0.185803	$< 10^{-14}$	$< 10^{-14}$	0.9938
3	0.5	−0.839538	−0.839538	$< 10^{-14}$	$< 10^{-14}$	0.9808
4	0	0.635548	0.636987	1.4×10^{-3}	1.8×10^{-4}	1
4	0.05	0.450582	0.453024	2.4×10^{-3}	3.0×10^{-4}	0.9988
4	0.2	−0.183485	−0.175303	8.2×10^{-3}	1.0×10^{-3}	0.9869
4	0.5	−1.691737	−1.659594	3.2×10^{-2}	3.7×10^{-3}	0.9605

IV. RESULTS WITH THE PARTICLE-HOLE TERM

A. Exact and variational references

Table II collects the reference energies, the exact ground-state energies and the UCCSD energies at $k = 3$ and $k = 4$. The $k = 4$ exact values reproduce Ref. [29] to all eight digits, now obtained in the full 70-dimensional $N = 4$ sector of the eight-qubit register rather than in the 36-dimensional $S_z = 0$ block (the ground state lies in that block for every f). As f grows the correlation energy grows much faster than the reference energy falls, from -0.36 at $f = 0$ to -1.69 at $f = 0.5$ (Fig. 1, left), the signature of a second, competing mean field. At $k = 3$ the UCCSD ansatz remains exact for every f, now with non-vanishing singles amplitudes (up to 0.09 at $f = 0.5$). At $k = 4$ the ansatz energy error grows by a factor of twenty across the scan, from 1.4×10^{-3} at $f = 0$ to 3.2×10^{-2} at $f = 0.5$, and the infidelity of the optimized state by the same factor, from 1.8×10^{-4} to 3.7×10^{-3}, tracking the weight of the exact ground state outside the seniority-zero space, which reaches 3.9% at $f = 0.5$ (Fig. 1, right). This is the first qualitative difference from the pairing model: the accuracy of the fixed single-reference ansatz, which did not depend on g before, now degrades with the strength of the pair-breaking channel. We note that these are the errors of the optimized Trotterized UCCSD circuit found by L-BFGS from a zero start; they document the accuracy of this preparation, not a proven expressivity limit of the ansatz. The UCCSD circuit itself is the same for every f — 26 operators, 1 312 CNOTs at $k = 4$ — since the particle-hole term changes the amplitudes, not the operator pool.

TABLE III. Refinement to $k = 4$ ($N = 4$, $g = 1$, $\mu = E_{\mathrm{low}} + 0.6$) with the particle-hole term: gap of H_{high}, minimum instantaneous gap along the path, initial overlap ov_0 of the prolonged state and adiabatic time T^* to reach overlap 0.99 (on the T grid of Sec. III A; the $T^* = 9$ quoted in Ref. [1] for the $3 \to 4$ chain at $f = 0$ used a finer grid), for the gate-free $2 \to 4$ chain and for the $3 \to 4$ chain started from the gate-level UCCSD state.

			$2 \to 4$		$3 \to 4$	
f	$\mathrm{gap}(H_{\mathrm{high}})$	ΔE_{min}	ov_0	T^*	ov_0	T^*
0	1.823	0.600	0.868	16	0.964	8
0.05	1.943	0.600	0.841	20	0.955	12
0.2	2.289	0.600	0.751	26	0.921	16
0.5	2.860	0.600	0.600	32	0.851	26

B. Refinement with pairs broken

We prolong the $k = 2$ state — at $N = 4$ still a single determinant, now with energy $E_{\mathrm{low}} = E_{\mathrm{ref}}$ — into the $k = 4$ space and evolve along the path of Eq. (3) with the same shift rule $\mu = E_{\mathrm{low}} + 0.6$, for each f (Fig. 2 and Table III). Two things change with f. The target state acquires seniority-two components that the prolonged determinant does not contain, and the effective pairing strength $g + 2f$ grows; both lower the initial overlap, from 0.868 at $f = 0$ to 0.600 at $f = 0.5$. The minimum gap along the path, however, is still set by the shift, $\Delta E_{\mathrm{min}} = 0.600$ for every f (the physical gap of H_{high} in fact *opens*, from 1.82 to 2.86), so the adiabatic time grows only because the state has further to travel: T^* rises from 16 to 32, and for every f the overlap exceeds 0.99 with the energy converging to FCI from above. The comparison with the coupling scan of Ref. [1] is telling: there, the pure pairing model at $g = 2$ — the value of $g + 2f$ at $f = 0.5$ — had $\mathrm{ov}_0 = 0.617$ and $T^* = 32$, almost exactly the numbers found here at $f = 0.5$. The refinement cost of the seniority-breaking model is thus governed by the renormalized pairing strength, and the genuinely new $1p$–$1h$ and pair-breaking channels add only a small correction to it. This is the same lesson as the coupling scan taught for the pairing force: the practical limiter of the method is not a closing gap but a degrading prolongation. The $k = 3 \to 4$ chain, started from the gate-level UCCSD state whose singles now carry non-zero angles, buys the degradation back: it starts at overlap 0.851 even at $f = 0.5$ and reaches 0.99 at $T^* = 26$ instead of 32. Whether that saves gates is decided, as in Ref. [1], by the conditioned coarse term of Eq. (4), which for the $3 \to 4$ chain is the complete $k = 3$ Hamiltonian — now 118 Pauli strings instead of 34 (Table I) — conditioned on the two appended qubits being empty; we have not compiled it with the particle-hole term and quote the $3 \to 4$ chain for its overlap only.

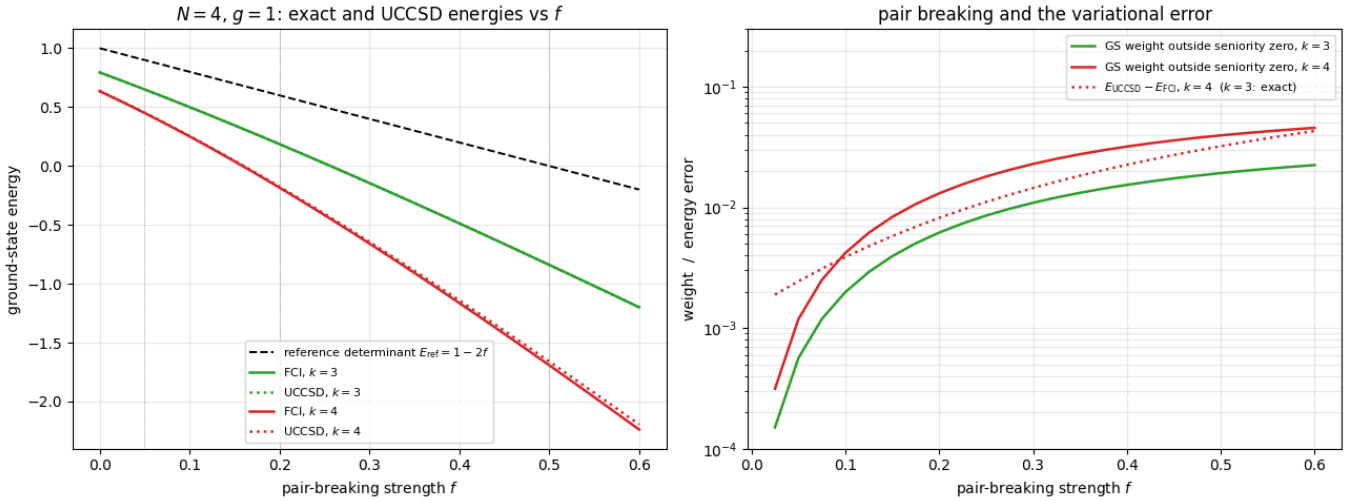


FIG. 1. The pairing model with the particle-hole term, $N = 4$, $g = 1$, as a function of the pair-breaking strength f. Left: energy of the reference determinant (dashed), exact ground-state energy (solid) and UCCSD energy (dotted) at $k = 3$ and $k = 4$; the vertical lines mark the strengths $f = 0.05, 0.2, 0.5$ of Table II. Right: weight of the exact ground state outside the seniority-zero subspace (solid) and the UCCSD energy error at $k = 4$ (dotted; at $k = 3$ the ansatz is exact).

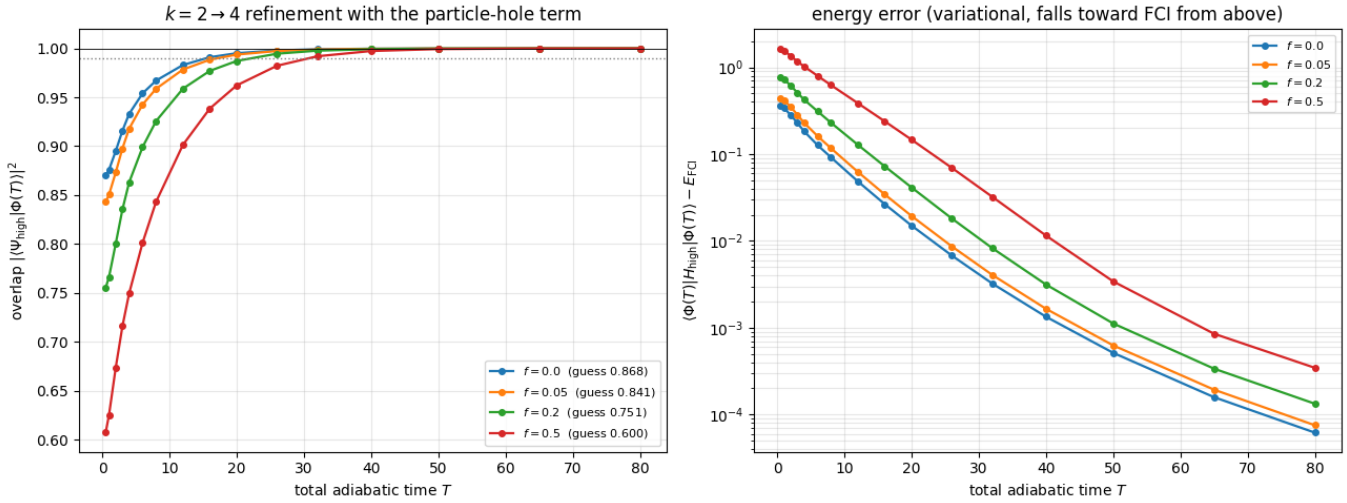


FIG. 2. Resolution refinement $k = 2 \to 4$ at $N = 4$, $g = 1$, for the pair-breaking strengths $f = 0, 0.05, 0.2, 0.5$. Left: overlap with the exact ground state versus the adiabatic time T (legend: initial overlap of the prolonged determinant; dotted line: the 0.99 target). Right: variational energy error $\langle\Phi(T)|H_{\rm high}|\Phi(T)\rangle - E_{\rm FCI}$ on a logarithmic scale.

C. Sufficient conditions versus practice

The conditions of Eq. (5) and the step condition that follows it can be evaluated exactly for the path of Eq. (3). Writing $A = P(H_{\rm low} - \mu)P^\dagger$ and $B = H_{\rm high} - \mu$, the schedule gives $\dot H = \frac{\pi}{2}\sin(\pi s)\,(B - A)$ and $\ddot H = \frac{\pi^2}{2}\cos(\pi s)\,(B - A)$, so the integrand of Eq. (5) is fixed by the single number $\|B - A\|$ and by the instantaneous gap $\Delta(s)$, both taken in the $N = 4$ sector in which the evolution takes place. Figure 3 shows the two along the $k = 2 \to 4$ path for the pure pairing force and the three particle-hole strengths, and Table IV collects the numbers; the $f = 0$ row is the reference of Sec. III C. Two opposite trends appear. The time bound hardly moves with f, $T_{\rm b} = 4.3$–5.1×10^3 for an infidelity of 10^{-2}: the minimum gap is set by the shift and is the same 0.600 for every f, the physical gap at $s = 1$ opens, and $\|B - A\|$ grows only from 7.62 to 8.42, so the integrand (which peaks near $s \simeq 0.25$, where $\sin(\pi s)$ is already sizable while the gap is still close to its minimum, and is dominated by the term $7\left\|\dot H\right\|^2/\Delta^3$) integrates to 435–511. Since the empirical T^* doubles over the same range, the overshoot of the bound falls from 3.2×10^2 at $f = 0$ to 1.4×10^2 at $f = 0.5$ — the particle-hole term brings

practice closer to the bound, not by making the bound tighter but by making the path genuinely longer. The step condition moves the other way. The commutator sum Λ_2 over the Trotter groups of the gate compilation (the conditioned coarse term, the diagonal term, the six pair-hopping terms and, for $f \neq 0$, the 48 particle-hole blocks of Sec. V) grows from 13 to 184, and with it the first-order step bound at T^* from 3.4×10^4 to 1.9×10^6, while the steps actually needed to reproduce the exact-slice overlap at T^* to within the adiabatic error of the path grow only from 160 to 640 (first order) and from 40 to 160 (second order). The bound charges the full norm of every new commutator, whereas the dynamics pays only for the matrix elements that couple the tracked state to its neighbors, and the small coefficient $f/2$ of the new blocks keeps the latter small. The ratios — 1.4–3.2×10^2 for the time and 2×10^2 to 3×10^3 for the step count — are the fermionic, seniority-mixed counterpart of what Ref. [34] reports for nucleation of a twelve-site Ising chain (6×10^1 and 10^7), and carry the same message: the conditions certify convergence but do not predict cost.

D. Rodeo filtering with a denser spectrum

The rodeo algorithm is the stage most exposed to the extra term, because its energy scan sees the whole spectrum of the eight-qubit Hamiltonian in the $N = 4$ sector, which the particle-hole term makes both denser and lower: 19 eigenvalues lie below $E = 4$ at $f = 0.5$ against 14 at $f = 0$. Figure 4 shows the energy scan ($M = 4$ cycles, 12 time sets per energy) at the strongest coupling, $f = 0.5$, feeding the filter the resolution-refined ($T = 25$) state with $p = 0.980$ and, for contrast, the prolonged determinant with $p = 0.600$. The coarse scan ($\sigma = 2$) resolves the ground state from the excited band — helped by the larger gap — and the fine scan ($\sigma = 8$) locates the peak at $E = -1.6924$ against the exact $E_0 = -1.691737$; with the refined input the excited-state structure that the prolonged input shows above the 2^{-M} background is essentially gone. Fixing $E = E_0$ and tracking the fidelity with the number of cycles for all four strengths (Fig. 5; exact controlled evolution, $\sigma = 3$, 40 Gaussian time sets), the infidelity falls geometrically along the $\frac{1-p}{p} 2^{-M}$ reference of Eq. (6) and the acceptance settles at the input overlap: after ten cycles $1 - \mathcal{F} = 2.0 \times 10^{-6}$, 2.9×10^{-6}, 4.6×10^{-6} and 4.5×10^{-5} for $f = 0$, 0.05, 0.2 and 0.5, with acceptances 0.998, 0.997, 0.994 and 0.980. The pair-breaking channels cost the pipeline one order of magnitude in final infidelity at the strongest coupling — recoverable with three more cycles — and nothing in acceptance beyond the lower input overlap. The denser spectrum does not slow the filter: the physical gap opens with f (Table III), so the smallest excitation energy carrying weight satisfies $\Delta_{\min}\sigma \geq 5$ at $\sigma = 3$ and the single-cycle suppression factor of Eq. (6) stays close to 1/2.

V. THE CIRCUIT-LEVEL BUDGET WITH THE PARTICLE-HOLE TERM

The pipeline itself is unchanged by $V_{\rm ph}$ — prolongation, the adiabatic path, the shift rule and the rodeo cycle were all written for the $2k$-qubit register — but the Hamiltonian that is Trotterized is not (Table I). To extend the circuit-level analysis of Ref. [1] we decompose $V_{\rm ph}$ into exactly exponentiable blocks in the same spirit as the pair-hopping terms. Its $q = r$ pieces are pairing terms and only rescale the rotation angles of the existing blocks ($g \to g + 2f$), at no cost. The remaining pieces, one per (p, q, r) with $q \neq r$, fall into two families: $2k(k-1)$ one-particle–one-hole blocks ($a^\dagger_{p\uparrow} n_{p\downarrow} a_{r\uparrow}$ + h.c. and the spin-flipped partner), each a sum of four Pauli strings, and $k(k-1)(k-2)$ pair-breaking blocks (eight strings each, like a double excitation). Within each block the strings mutually commute, so every factor $e^{-i\hat{T}\delta t}$ is realized exactly by its rotations and the product-formula error again comes only from the non-commutativity *between* blocks: 55 of them at $k = 4$ instead of seven.

Two things follow, both quantified in Fig. 6 and Table V. First, the Trotter error at fixed step count grows only modestly — at $t = 1$, $k = 4$ the second-order error with sixteen steps is 2.1×10^{-3}, 3.8×10^{-3} and 7.5×10^{-3} for $f = 0$, 0.2, 0.5 — because the new blocks carry the small coefficient $f/2$. The Trotterized refinement reaches its exact-slice value with the same $n_s = 80$ second-order steps at $f = 0.2$ (overlap 0.9961 against 0.9969), and the rodeo cycle at $\delta t = 0.25$ keeps an acceptance of 0.96 with an infidelity of 3.8×10^{-4} after six cycles, essentially the pairing-model operating point. At $f = 0.5$ the step $\delta t = 0.25$ begins to fail — acceptance 0.84, the symptom of a corrupted evolution recalled in Sec. III C — and $\delta t = 0.125$ restores it to 0.97 with an infidelity of 6×10^{-4} against 5×10^{-4} for exact evolution. The filter is thus again diagnostic of its own Trotter error, and the diagnostic is reached at a step size that was safe for the pairing force; since the number and structure of the Trotter blocks are the same at $f = 0.2$ and $f = 0.5$, it is the strength of the new channels, not their number, that shrinks the admissible step. Second, the depth of one step grows ninefold: compiled block by block with the ladder rule of Eq. (7), a first-order step costs 2728 CNOTs instead of 296 (348 rotations instead of 60; 3424 against 416 once controlled), the $1p$–$1h$ family contributing 640 and the pair-breaking family 1792 CNOTs; the coarse term is unchanged (14 CNOTs, the coarse space is still one determinant). The budget per accepted preparation therefore scales almost exactly with the step cost: at the pairing-model operating point ($n_s = 80$, $M = 2$, $\delta t = 0.25$) it is 6.5×10^5 CNOTs at $f = 0.2$ against 7.2×10^4 for the pure pairing force, for an infidelity of 1.0×10^{-3}, and at $f = 0.5$ the finer rodeo step brings the total to 8.3×10^5 CNOTs for 5.3×10^{-3}. The price of a more realistic interaction thus appears where it should, in the depth of each Trotter step of the Hamiltonian evo-

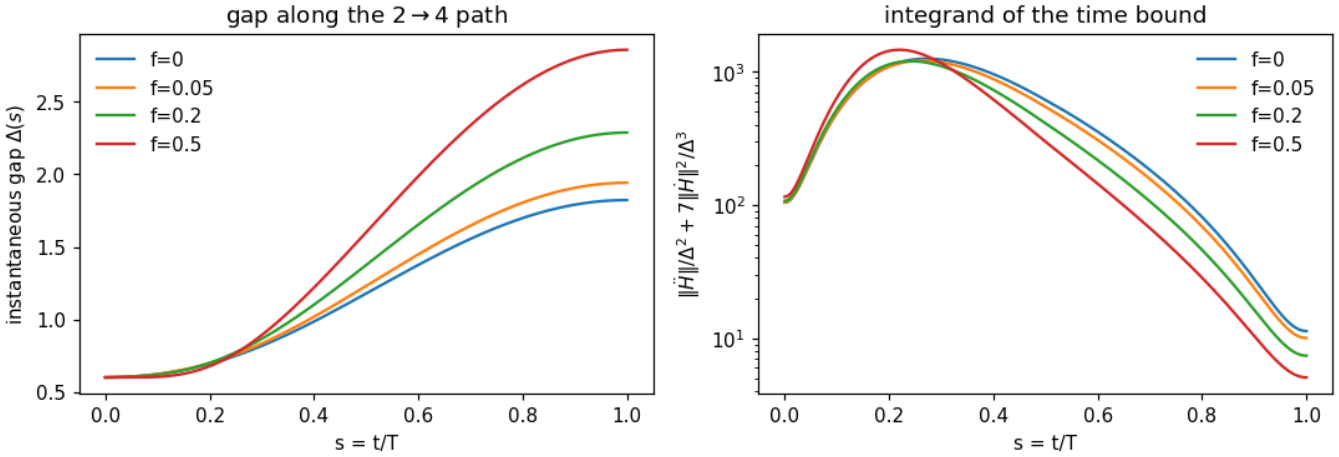


FIG. 3. The ingredients of the adiabatic-time bound, Eq. (5), along the $k = 2 \to 4$ refinement path at $N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$, for the pure pairing model and the three particle-hole strengths. Left: instantaneous gap $\Delta(s)$ in the $N = 4$ sector, from the shift-controlled 0.600 at $s = 0$ to the physical gap of $H_{\rm high}$ at $s = 1$, which opens with f. Right: the integrand $\left\|\ddot{H}\right\|/\Delta^2 + 7\left\|\dot{H}\right\|^2/\Delta^3$ (logarithmic scale), which peaks near $s \simeq 0.25$ and integrates to 435–511 (Table IV).

TABLE IV. Sufficient conditions of Ref. [34] evaluated for the $k = 2 \to 4$ refinement path ($N = 4$, $g = 1$, $\mu = E_{\rm low} + 0.6$; spectral norms in the $N = 4$ sector) against the empirical cost, for the pure pairing model ($f = 0$, the reference of Ref. [1]) and the three particle-hole strengths. $\Delta_{\rm min}$: minimum instantaneous gap; I: the integral in Eq. (5); $T_{\rm b}$: the time bound for infidelity $\epsilon = 10^{-2}$, $T_{\rm b} = I/\sqrt{\epsilon}$; T^*: time for overlap 0.99 (grid of Sec. III A); $\Lambda_1 = \max_s \left\|\dot{H}\right\|$ and $\Lambda_2 = \max_s \sum_{i<j} \|[H_i, H_j]\|$ over the Trotter groups of the gate compilation; $n_{\rm b}(T^*)$: first-order Trotter steps required by the bound along the T^* path at $\eta = \sqrt{\epsilon}/2$; n_1, n_2: first- and second-order steps (from the sequence $5, 10, 20, \ldots$) that reproduce the exact-slice overlap at T^* to within the adiabatic error $1 - p(T^*)$ of the path.

f	$\Delta_{\rm min}$	$\|B - A\|$	I	$T_{\rm b}$	T^*	$T_{\rm b}/T^*$	Λ_1	Λ_2	$n_{\rm b}(T^*)$	n_1	n_2
0	0.600	7.62	511	5.1×10^3	16	3.2×10^2	12.0	13.0	3.4×10^4	160	40
0.05	0.600	7.66	480	4.8×10^3	20	2.4×10^2	12.0	21.6	8.8×10^4	320	40
0.2	0.600	7.85	435	4.3×10^3	26	1.7×10^2	12.3	58.7	4.0×10^5	320	80
0.5	0.600	8.42	446	4.5×10^3	32	1.4×10^2	13.2	184	1.9×10^6	640	160

lution, and not in the structure, the operating point or the convergence of the pipeline.

Measured against the right comparator, however, the extra depth buys no accuracy advantage. The actual infidelity of the UCCSD state is 1.0×10^{-3} at $f = 0.2$ and 3.7×10^{-3} at $f = 0.5$ (Table II), so the hybrid at $M = 2$ is tied with the *unfiltered* variational state at $f = 0.2$ and behind it at $f = 0.5$; and the same filter applied to the UCCSD input (last column of Table V) reaches 5.1×10^{-4} and 2.0×10^{-3} at $\delta t = 0.25$, 2.6×10^{-4} and 9.6×10^{-4} at $\delta t = 0.125$, for 2.0–3.8×10^5 CNOTs per accepted preparation — a factor two to three below the hybrid in cost and two to five below it in infidelity. The reason is the one identified in Sec. IV B: the variational input degrades with f, but so does the prolonged starting state of the refinement, from $p = 0.868$ to 0.600, so the refinement must travel further — T^* doubles — along a path whose every step has become nine times deeper. The seniority-breaking term thus confirms the structural robustness of the pipeline and, at the same time, the negative conclusion of Ref. [1].

VI. DISCUSSION AND OUTLOOK

Opening the particle-hole channel changes the Hamiltonian in exactly the ways that make the pure pairing model a special case — the seniority is mixed, the spectrum is denser, the reference determinant is no longer self-consistent, singles switch on in the variational ansatz, and the compact pair-to-qubit encoding is gone — and leaves the pipeline untouched. The same prolongation, the same shift rule, the same adiabatic path and the same filter run on the eight-qubit register without any modification, the path gap stays at the shift-controlled 0.600 for every strength, the refinement time doubles for the same reason it grew in the coupling scan of the pairing force — a poorer prolonged start, governed here by the renormalized strength $g + 2f$ — and the rodeo filter converges geometrically to 10^{-5}–10^{-6} in ten cycles with the accep-

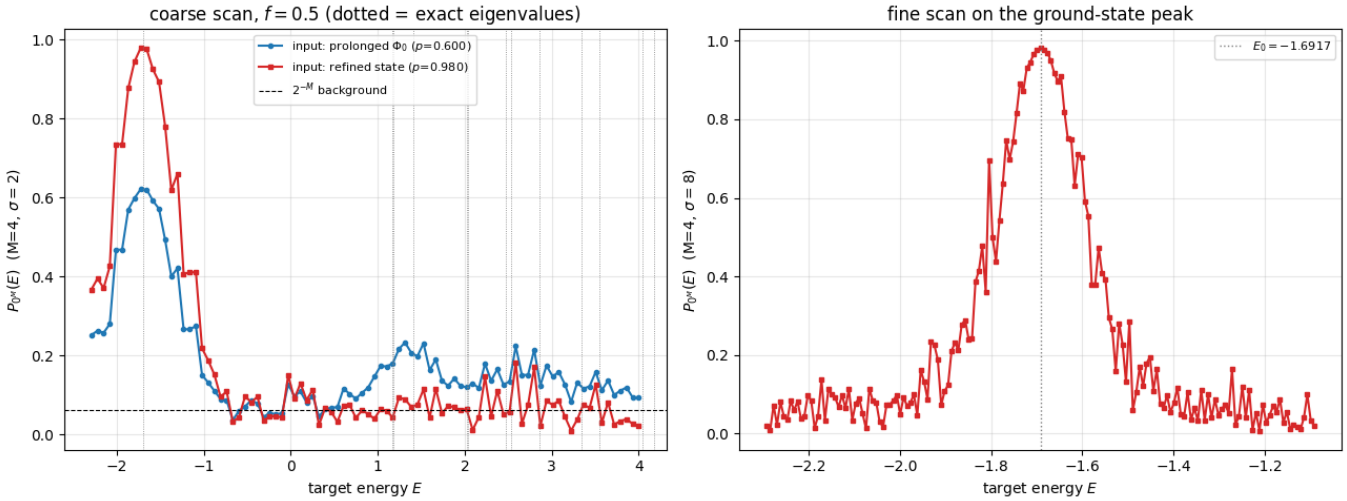


FIG. 4. Rodeo energy scan ($M = 4$) for the pairing model with the particle-hole term at $f = 0.5$. Left: coarse scan ($\sigma = 2$) from the prolonged determinant ($p = 0.600$) and from the resolution-refined state ($p = 0.980$); dotted vertical lines mark the exact eigenvalues, the dashed line the 2^{-M} background. Right: fine scan ($\sigma = 8$) on the ground-state peak, located at $E = -1.6924$ (exact $E_0 = -1.6917$).

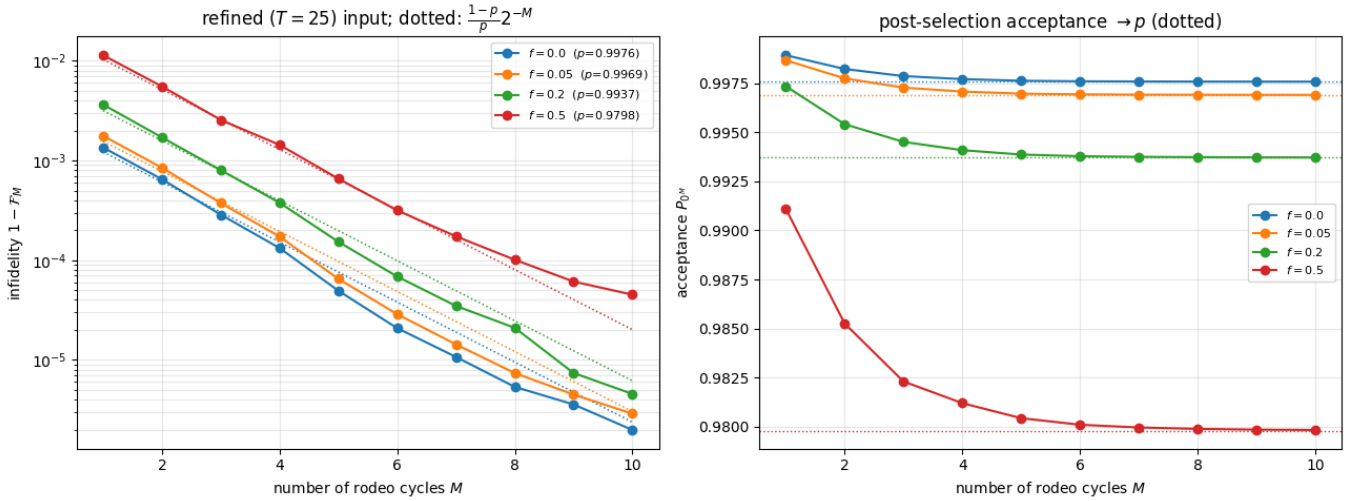


FIG. 5. Rodeo eigenstate preparation at $E = E_0$ with the particle-hole term, for $f = 0, 0.05, 0.2, 0.5$, from the resolution-refined ($T = 25$) input, averaged over 40 random time sets ($\sigma = 3$). Left: post-selected infidelity versus the number of cycles M (dotted: $\frac{1-p}{p}2^{-M}$). Right: the acceptance converges to the input overlap p (dotted).

tance at the input overlap. The sufficient conditions of Ref. [34] are consistent with this picture from the other side: the time bound is insensitive to the pair breaking because the minimum gap is set by the shift, while the step bound, which charges the full norm of every new commutator, grows by almost two orders of magnitude where practice grows by a factor four. Everything the particle-hole term costs is in the depth of a Trotter step, 296 → 2728 CNOTs, traceable string by string to Table I, and the budget per accepted preparation scales with it, from 7×10^4 to 6–8 $\times 10^5$ CNOTs.

The question that motivated the extension — whether the refinement stage begins to earn its gates once the single-reference variational input starts to fail — receives, at these strengths, a negative answer, and the reason is instructive. The UCCSD infidelity does degrade twenty-fold across the scan, from 1.8×10^{-4} to 3.7×10^{-3}, as the exact ground state acquires up to 3.9% seniority-two weight that a fixed single-reference ansatz from an unrelaxed determinant captures imperfectly. But the prolonged starting state of the refinement, which is the same determinant, degrades as well — from $p = 0.868$ to 0.600, with T^* doubling — so the adiabatic stage has further to travel along a path whose every step has become nine times deeper, and at the operating point the hybrid is tied with the unfiltered UCCSD state at $f = 0.2$, behind

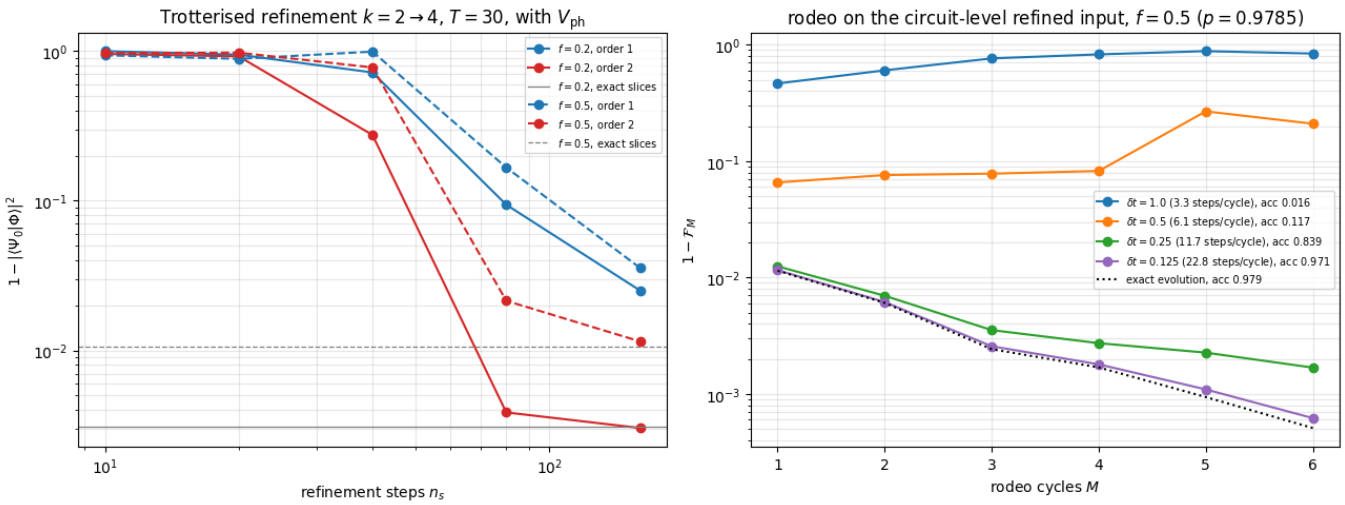


FIG. 6. The pipeline at the circuit level with the particle-hole term ($k = 4$, $N = 4$, $g = 1$). Left: infidelity of the Trotterized refinement $k = 2 \to 4$ ($T = 30$) versus the number of product-formula steps n_s for first- and second-order formulas at $f = 0.2$ (solid) and $f = 0.5$ (dashed); horizontal lines mark the exact-slice refinement. Right: rodeo infidelity versus cycles at $f = 0.5$ on the circuit-level refined input ($n_s = 80$, second order, $p = 0.9785$) with the controlled evolution Trotterized at fixed step sizes δt; the legend gives the mean number of steps per cycle and the six-cycle acceptance.

TABLE V. Gate budget at $k = 4$ ($N = 4$, $g = 1$) with the particle-hole term, per accepted preparation, at the operating point of Sec. III B ($n_s = 80$ second-order refinement steps at $T = 30$, $M = 2$ rodeo cycles, $\sigma = 4$) for two rodeo step sizes δt. Step costs are the block-by-block CNOT counts (uncontrolled / controlled); the refinement column includes the 14-CNOT coarse term per half-step; the rodeo column uses the mean number of Trotter steps over the two cycles, doubled for the symmetric formula. The last two columns give the infidelity of the hybrid and, for comparison, the infidelity of the unfiltered UCCSD state (1 312 CNOTs, 26 parameters, the same circuit for every f) and of the UCCSD state passed through the same two-cycle filter, with its CNOTs per accepted preparation in parentheses. The first row is the pure pairing model of Ref. [1].

f	δt	step (unctrl./ctrl.)	refinement	rodeo	per run	P_M	per accepted	$1 - \mathcal{F}$: hybrid / UCCSD	UCCSD+rodeo (CNOTs)
0	0.25	296 / 416	49 600	22 000	71 600	0.994	7.2×10^4	5.3×10^{-4} / 1.8×10^{-4}	1.3×10^{-4} (2.3×10^4)
0.2	0.25	2728 / 3424	438 720	190 603	629 323	0.972	6.5×10^5	1.0×10^{-3} / 1.0×10^{-3}	5.1×10^{-4} (2.0×10^5)
0.2	0.125	2728 / 3424	438 720	375 499	814 219	0.996	8.2×10^5	8.1×10^{-4} / 1.0×10^{-3}	2.6×10^{-4} (3.8×10^5)
0.5	0.25	2728 / 3424	438 720	190 603	629 323	0.891	7.1×10^5	6.1×10^{-3} / 3.7×10^{-3}	2.0×10^{-3} (2.1×10^5)
0.5	0.125	2728 / 3424	438 720	375 499	814 219	0.979	8.3×10^5	5.3×10^{-3} / 3.7×10^{-3}	9.6×10^{-4} (3.8×10^5)

it at $f = 0.5$, and behind the filtered UCCSD state by a factor two to three in cost and two to five in infidelity at both strengths compiled (Table V). A regime in which refinement wins must therefore combine a poor full-space variational input with a good coarse one, so that the prolonged overlap stays high while the fine-space ansatz fails; the $3 \to 4$ chain, which starts at overlap 0.851 even at $f = 0.5$, is the natural candidate, but it carries the conditioned $k = 3$ Hamiltonian — now 118 Pauli strings — along the whole path, and in Ref. [1] that coarse term already cost more per application than the fine-space step itself. Whether the balance tips with the particle-hole term is a compilation we have not carried out, and we list it below. Two further caveats apply to the comparator. The reference determinant was not re-optimized, so part of what the singles absorb is mean-field relaxation; a Hartree–Fock-optimized reference would lower the variational error at $f \neq 0$ and, by the same token, raise the prolonged overlap of a refinement started from it, so the comparison is between two unrelaxed starts and should be repeated with relaxed ones. And the variational numbers are those of one optimizer from one start, which document the accuracy of this preparation rather than the expressivity of the ansatz.

a. Encodings and the pairing-model literature. The seniority-breaking term is also the point at which the present work and the pairing-model programme of Lacroix and collaborators part company at the level of the register. The symmetry-restoration route [15–19], the ADAPT-VQE studies [10, 21] and the Green's-function construction of Ref. [22] are formulated for the seniority-conserving pairing force, for which one qubit per level suffices and the $N = 4$, $k = 4$ problem is a six-dimensional block; with V_{ph} that block becomes the 36-dimensional $S_z = 0$ sector of a fermionic register, and every gate count of Sec. V refers to the $2k$-qubit Jordan–Wigner en-

coding on which it is written. The structural relation between the two approaches is unchanged by the term: both start from a cheap, imperfect trial state and sharpen it, the symmetry-restoration route by projecting on a symmetry quantum number and ours by filtering in energy along a gapped path across resolutions, the two filters being the same resource — one ancilla, a controlled unitary and a post-selected measurement — that compose. What the particle-hole term adds is the observation of Ref. [10] seen from the other side: there, ADAPT-VQE with a symmetry-breaking pool could collapse to unphysical energies on degenerate pairing systems, and a downstream energy filter cannot repair an input of zero overlap or separate degenerate states by energy alone; here, the Hamiltonian itself breaks the seniority, the singles and pair-breaking doubles of a symmetry-conserving pool are physical, and the filter has nothing to repair but the adiabatic tail. A seniority-mixed Hamiltonian is, in that sense, the natural target for an ADAPT-style input with a full singles-and-doubles pool, and its inclusion as a comparator is the obvious next step.

b. Hardware requirements. The Trotterized pipeline quantifies the dominant coherent error of an ideal device, and the requirements with the particle-hole term are explicit: $2k+1 = 9$ qubits at $k = 4$; a mid-circuit measurement with reset and reuse of one ancilla; arbitrary-angle R_z rotations, 160 per UCCSD preparation and 348 per Trotter step (against 60 for the pairing force); and a coherent budget per accepted preparation of 2–4×10^5 two-qubit gates for the cheapest route of Table V (UCCSD input, two cycles) and 6–8×10^5 for the refined input, on top of the energy scan and the repetitions needed to estimate acceptance probabilities. At the two-qubit error rates demonstrated in large-scale experiments today — of order 10^{-2} on the superconducting processor of Ref. [36] and $(0.8\text{–}2) \times 10^{-3}$ on trapped-ion processors [37, 38] — a 10^5-gate run suffers of order one hundred faults even on the best of these, and the full particle-hole budget is nearly an order of magnitude beyond that; none of it is executable with physical qubits without substantial error mitigation, and it is a workload for the first generation of logical-qubit machines, such as the neutral-atom systems on which logical circuits with mid-circuit readout have been demonstrated [39, 40] and the machine planned by the QuNorth initiative [41]; Table V is offered as its specification. Connectivity matters more than for the pairing force: all-to-all connectivity absorbs the Jordan–Wigner ladders without SWAP overhead, and on square or heavy-hex lattices the weight-4 pair-hopping rotations act on adjacent qubits, but the weight-$2k$ strings of the $1p$–$1h$ family do not, so routing costs are no longer bounded by the pair structure. A proportionate near-term target remains a mitigated $k = 3$ demonstration on seven qubits, where the Hamiltonian has 118 strings of weight up to six, with a short refinement schedule and one or two rodeo cycles, and where the collapse of the acceptance under a corrupted evolution — reached here already at $\delta t = 0.25$, $f = 0.5$ — doubles as a hardware diagnostic.

c. Concluding remarks and perspectives. The pairing model with a particle-hole interaction is the smallest Hamiltonian that has the central ingredients of a realistic two-body interaction — pair scattering, one-particle–one-hole excitations and pair breaking — while remaining exactly solvable in a 70-dimensional sector against which every stage of a quantum state-preparation pipeline can be validated, and the present results establish how resolution refinement followed by the rodeo algorithm behaves on it: unchanged in structure and convergence, nine times deeper per Trotter step, and still not a gate saving over filtering a cheap variational input, which the term degrades but does not break. Four extensions follow directly. First, the comparators that the seniority-mixed Hamiltonian calls for: a Hartree–Fock-relaxed reference for both the variational ansatz and the prolonged start, an ADAPT-VQE input with a full singles-and-doubles pool, and the compiled $3 \to 4$ chain with the conditioned $k = 3$ particle-hole Hamiltonian, which is where a regime favorable to refinement would first appear. Second, an optimized refinement in the sense of the optimized nucleation of Ref. [34] — the Trotterized path of Eq. (3) as a warm-started variational circuit with a few layers per stage — for which the ninefold step depth of the particle-hole model is precisely the motivation. Third, other seniority-breaking Hamiltonians on which the same refinement-plus-filter analysis can be repeated against exact diagonalization, the Hubbard model on small lattices being the obvious next case, since there basis refinement and the lattice growth of nucleation can be composed with the same Trotterized machinery. Fourth, a mitigated seven-qubit demonstration on present devices, followed by the nine-qubit budget of Table V on an early logical-qubit machine.

CODE AVAILABILITY

The complete implementation — pairing Hamiltonian with and without the particle-hole term, gate-level UCCSD–VQE, prolongation, adiabatic refinement, and the ancilla-based rodeo algorithm — is provided as a tested Python package and a single Jupyter notebook, shared with Ref. [1], that reproduces every figure and number in this article. Give GitHub link when everything is ready.

ACKNOWLEDGMENTS

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